

Comparable Elements in Integer Lattices

GREEN-054

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Consider a set $S \subset [N]^3$ with the property that any two distinct elements s, s' of S are comparable (in the coordinatewise partial order). Is $|S| \leq N^{2-\delta}$ for some $\delta > 0$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>